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TerraCrate

Secure, offline-first file storage for local networks

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Value Proposition

Existing file-sharing solutions force a trade-off between security and portability. USB drives lack access control, commercial NAS requires permanent infrastructure, and cloud platforms introduce third-party trust dependencies and internet requirements. No lightweight solution combines certificate-based authentication, fine-grained access control, and zero-configuration discovery in a single self-contained device.

Objective

Build a portable, self-hosted file server on a Raspberry Pi that provides secure multi-user file sharing with mTLS authentication, three-tier permissions, and mDNS discovery, deployable in any environment without cloud infrastructure or client software beyond a browser.

Problem Definition

TerraCrate gives you full custody of your data in a device you can carry in your pocket. Plug it in and it is discoverable. Approve a user and they get a certificate in their inbox. Pull the plug and your data goes with you. No cloud, no residue, no trust required.

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